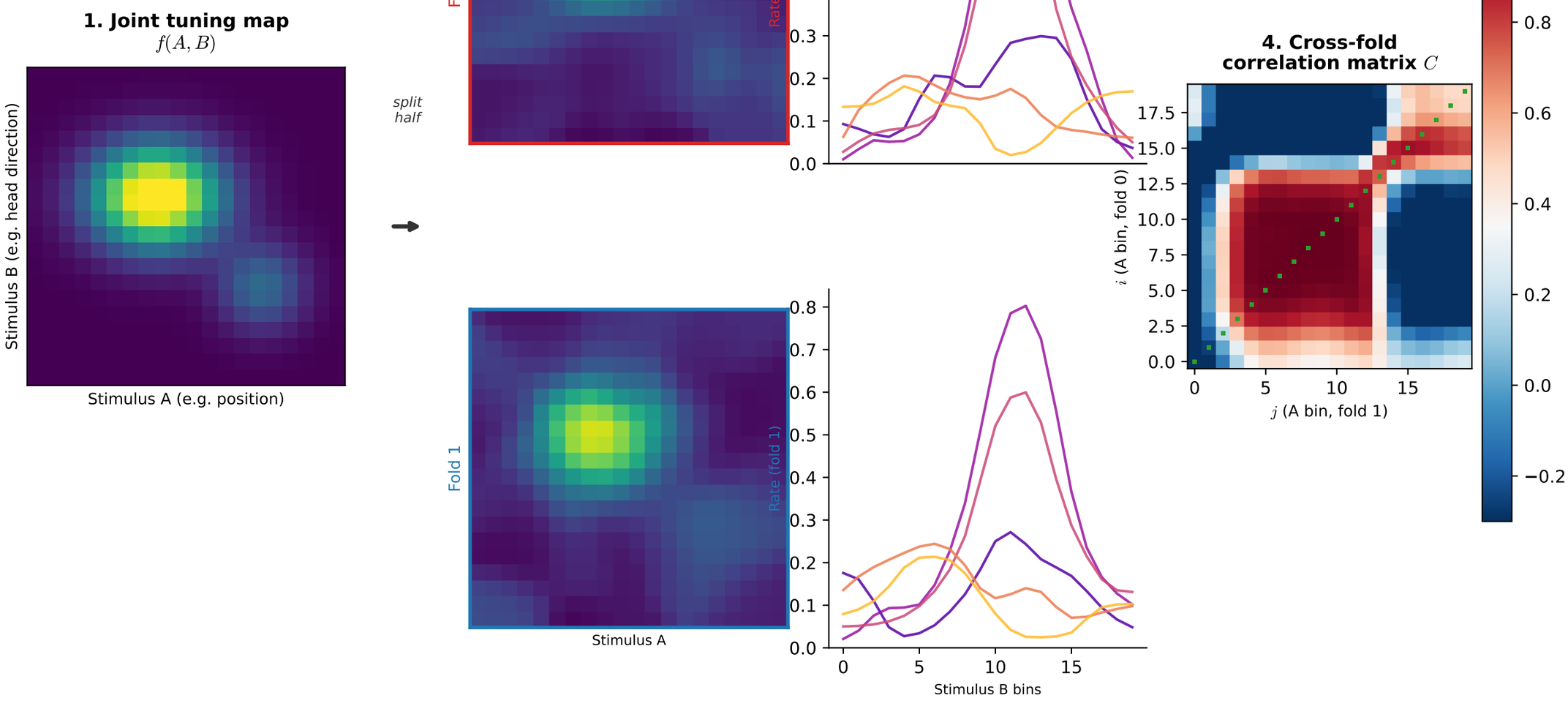
Cross-Fold Correlation Method for Decomposing Stimulus Contributions




```
5. Read off the matrix
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Diagonal (green dots):
  mean r = 0.689
  = Reliability
  (same-bin cross-fold r)

Off-diagonal mean:
  mean r = 0.147
  = Similarity of tuning
    across A positions

Reported EV(A):
  1 - off-diag = 0.853

High EV(A):
  off-diag << diag
  -> A modulates firing

Low EV(A):
  off-diag ~ diag
  -> A has little effect
```

Step 3: For each bin a_i of stimulus A, extract the tuning curve over B: $\mathbf{f}_i^{(0)} = f_{\text{fold}\,0}(a_i,\ \cdot\ )$ and $\mathbf{f}_j^{(1)} = f_{\text{fold}\,1}(a_j,\ \cdot\ )$

Step 4: Cross-fold correlation matrix: $C_{ij} = \text{corr}\left(\mathbf{f}_i^{(0)},\ \mathbf{f}_j^{(1)}\right)$

Step 5: $\text{Reliability} = \langle C_{ii} \rangle_i \qquad \text{EV}(A) = 1 - \langle C_{ij} \rangle_{i \neq j}$